

Rapid Microstructure Prediction in Laser Powder Bed Fusion Using Bayesian Updating of Eagar–Tsai Model

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Core Thesis

Composition



- A 5-element affine projection represents the accessible composition simplex.
- Interior coloring encodes local element content across the design space.
- Bayesian updates narrow the search to feasible LPBF compositions.

Solidification Condition

- Process parameters determine thermal gradients, cooling rates, and melt-pool shape.
- The Eagar–Tsai model provides fast temperature-field priors across the parameter space.
- These fields define the local solidification path for each candidate build condition.

Solidified Microstructure

- Microstructure emerges from the coupling of chemistry and thermal history.
- Predicted G/R conditions map to expected grain morphology and phase selection.
- The workflow targets rapid microstructure prediction from composition to final structure.



Motivation and Constraints

- LPBF alloy design spans large composition–processing spaces.
- Resulting microstructures depend on thermal gradients, G , and solidification rates, R .
- Fast thermal models are needed for high-throughput screening before expensive simulations or experiments.

Key constraint

The Eagar–Tsai model is fast enough for screening, but its simplified physics limits fidelity:

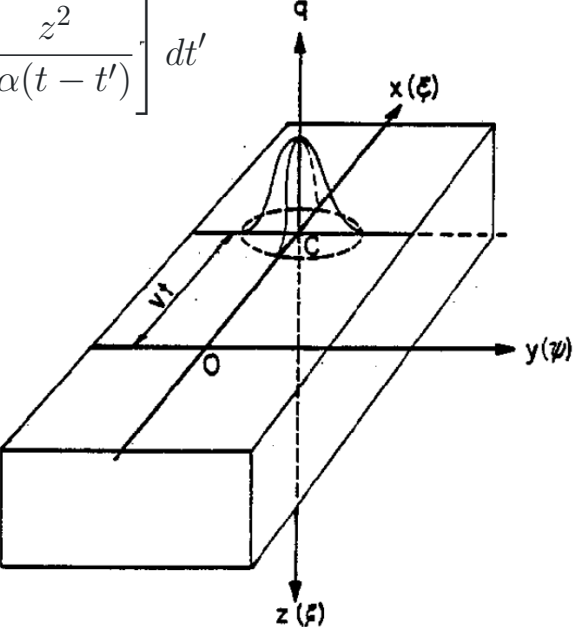
- temperature-invariant properties,
- analytical melt-pool assumptions,
- reduced accuracy relative to FEM thermal simulations.



Eagar-Tsai Model Foundation

Original Eagar-Tsai Paper

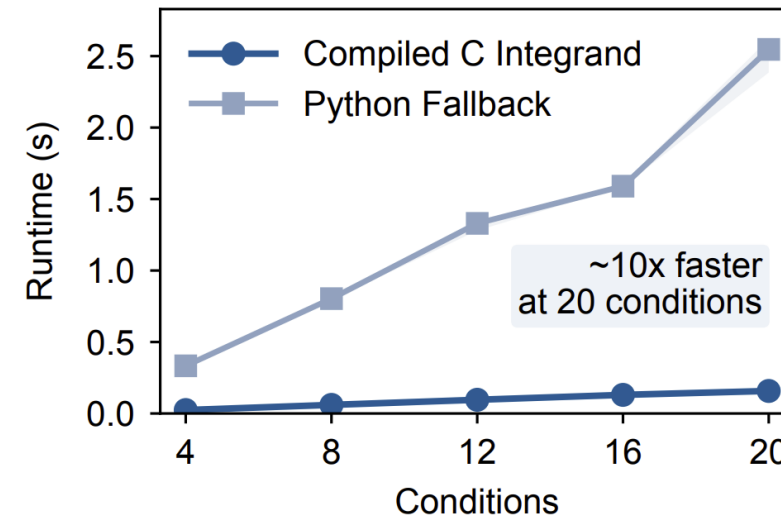
Transient Gaussian moving heat source

$$\begin{aligned} T - T_0 &= \frac{1}{2} \int_0^t dT_i \\ &= \frac{q}{\pi \rho C_p \sqrt{4\pi\alpha}} \int_0^t \frac{(t-t')^{-1/2}}{2\alpha(t-t') + \sigma^2} \\ &\quad \times \exp \left[-\frac{(x-vt')^2 + y^2}{4\alpha(t-t') + 2\sigma^2} - \frac{z^2}{4\alpha(t-t')} \right] dt' \end{aligned}$$


Eagar, T. W., & Tsai, N. S. (1983). Temperature fields produced by traveling distributed heat sources. *Welding Journal*, 62(12), 346-355.

Refactored Eagar-Tsai Model

Python library implementation with compiled-C speedup



- Uses the Rubenchik nondimensional integrand reformulation.
- Compiled C integrand removes Python overhead inside QUADPACK.
- Adds batch APIs, plotting, domain expansion, and field export.



Refactored ET Model

Core idea

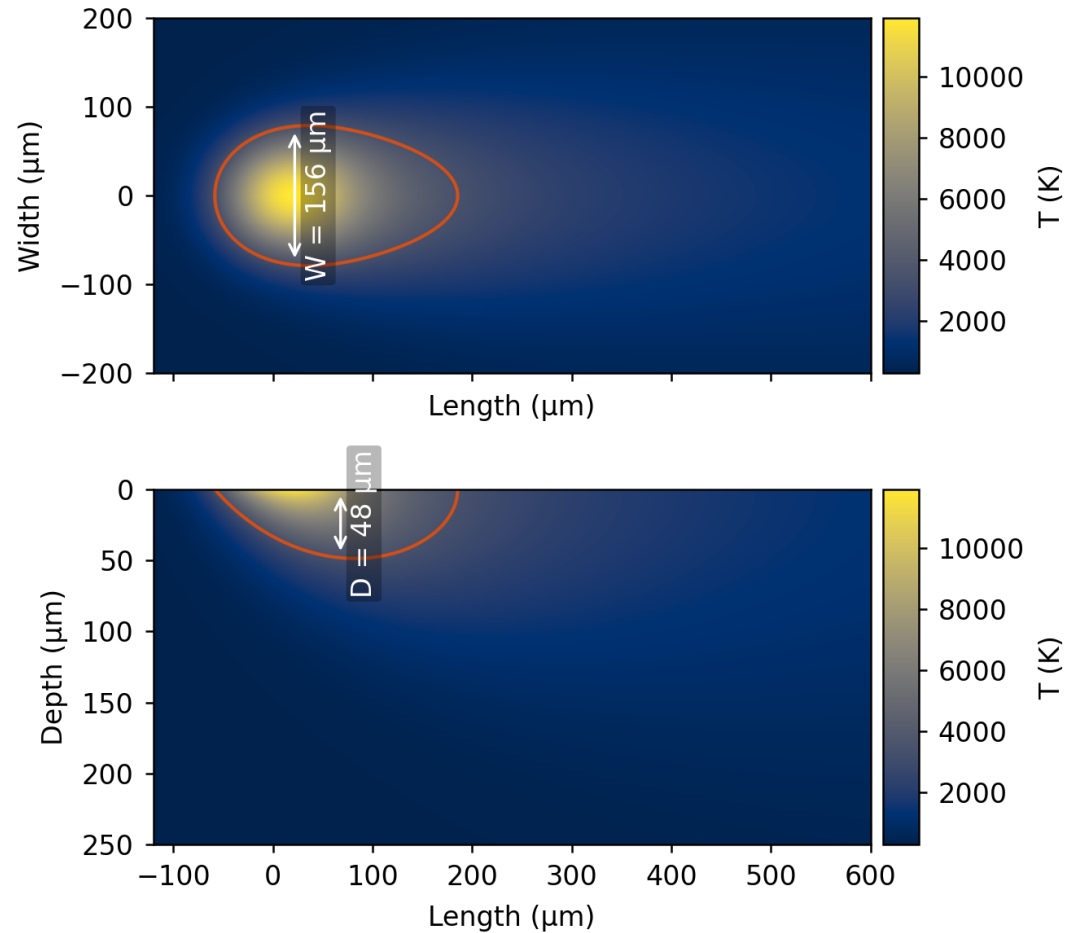
The refactored Eagar–Tsai workflow separates melt-pool prediction, 2D heatmap generation, and 3D temperature-field export into reusable steps.

- Library-backed execution supports direct melt-pool evaluation from process and material inputs.
- Temperature fields can be exported as both dense CSV volumes and VTI files for downstream analysis.
- The workflow is structured for Bayesian correction, microstructure inference, and repeatable parameter studies.

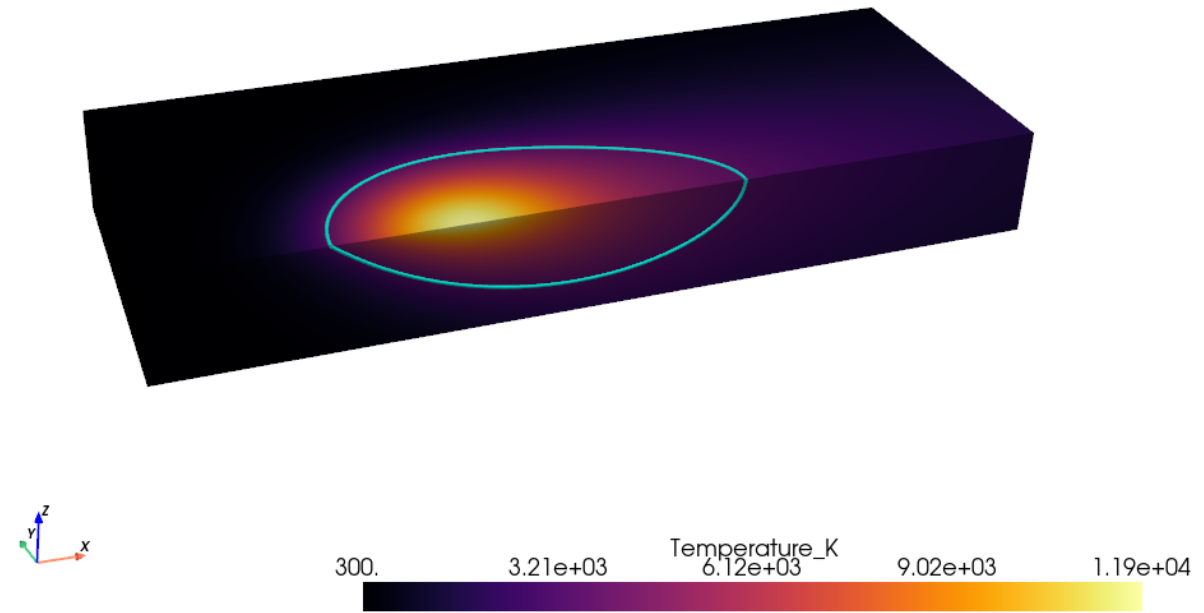
Note to self: add speedup results and refer to Dogu's JOS paper.



ET Thermal Cross Sections

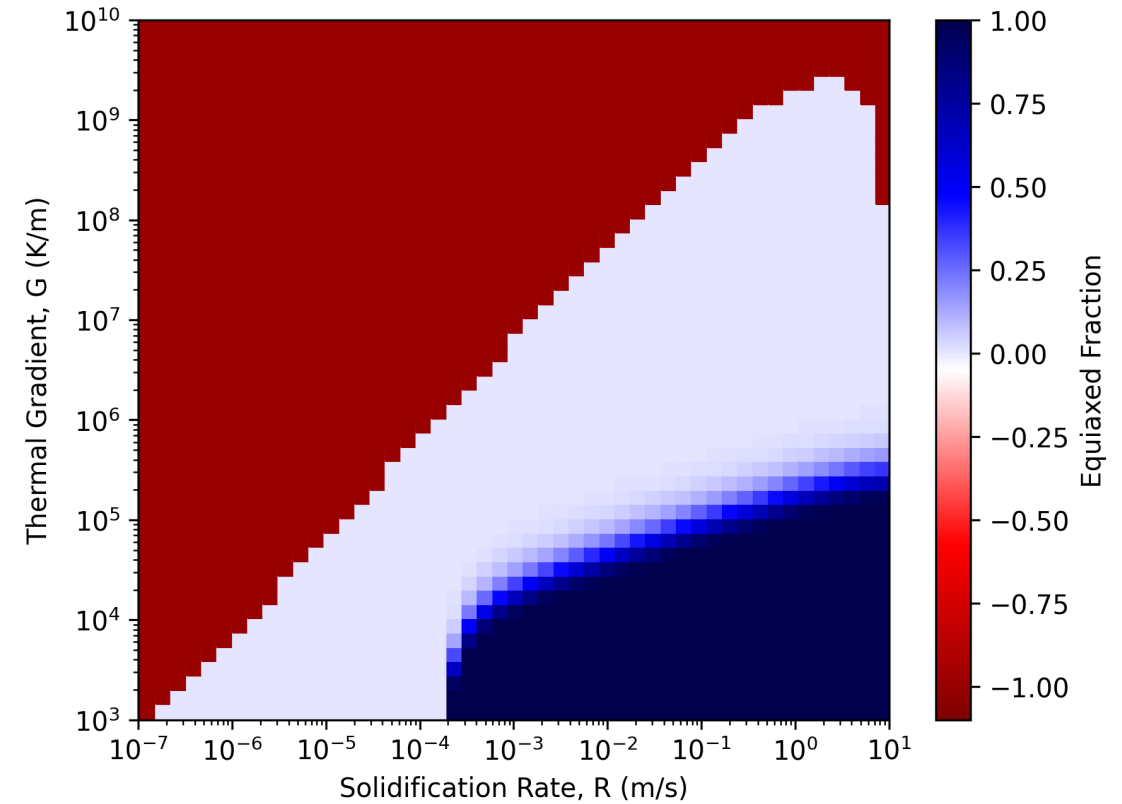
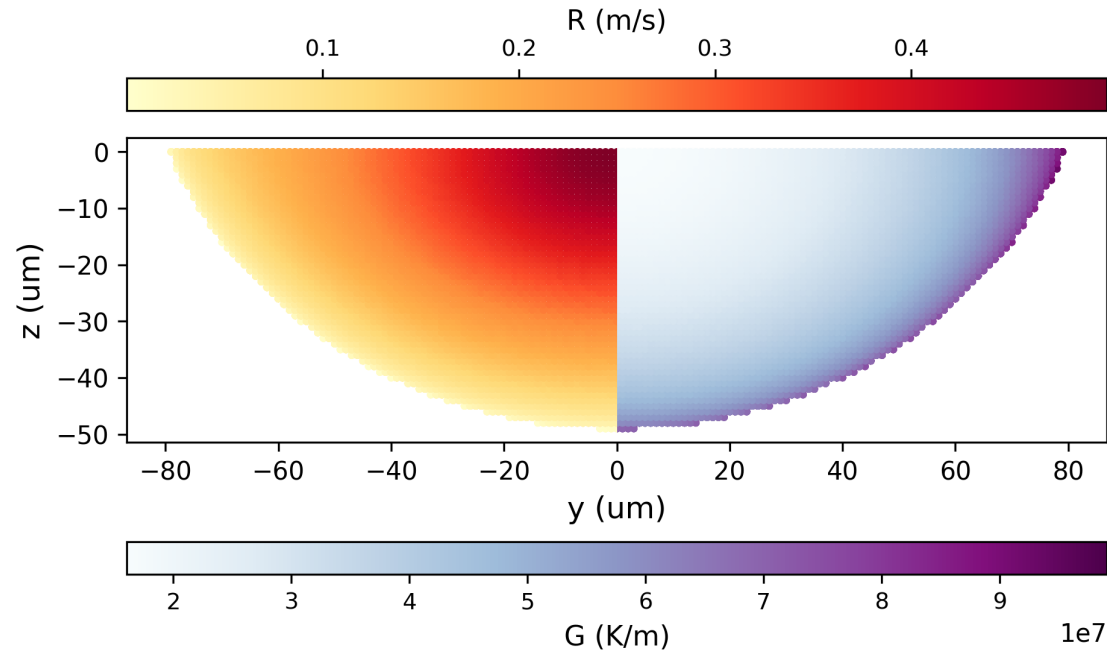


Temperature heatmap for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250 \text{ W}$ and $V = 0.5 \text{ m/s}$. Top: $z = 0$ surface. Bottom: $y = 0$ cross section.



3D view of the ET melt pool for the same alloy and process condition.

ET Thermal Gradient and Solidification Rate

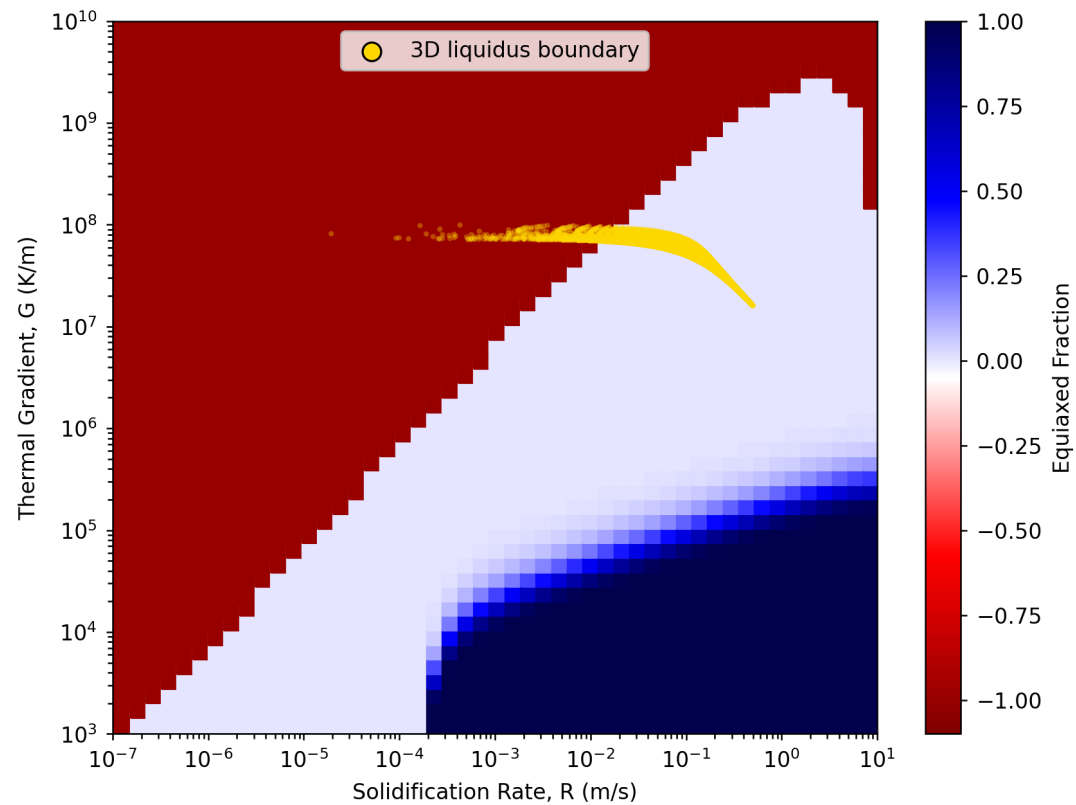


Projection of G and R along the solidification front onto a yz slice for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

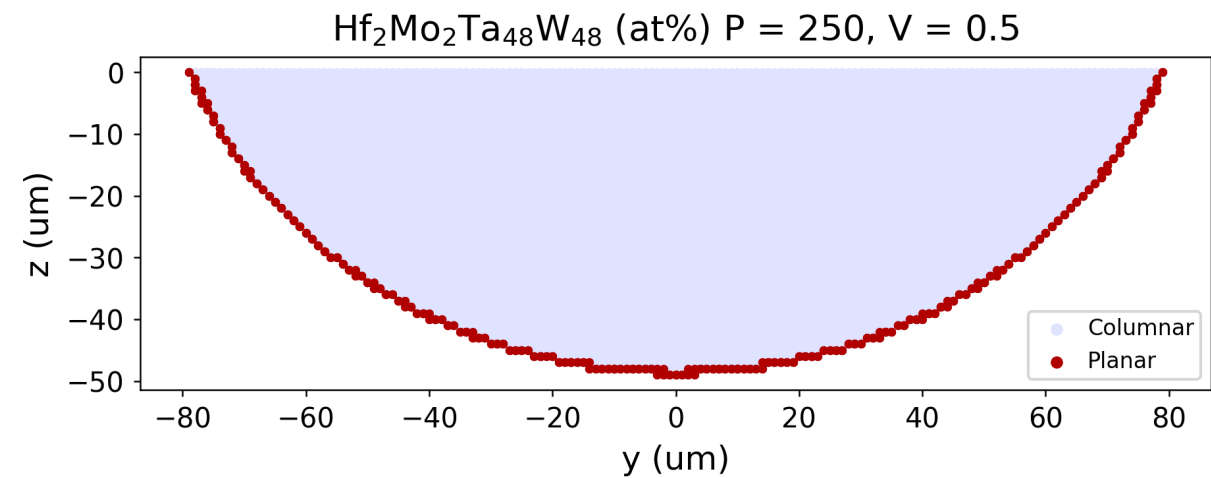
GR Map calculated with Thermo-Calc CET Property Model for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.



ET Microstructure Classification



GR Map calculated with Thermo-Calc CET Property Model for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.



Projected microstructure classification along the solidification front onto a yz slice for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

Thermo-Calc AM Module



TCAM Thermal Cross Sections



TCAM Thermal Gradient and Solidification Rate



TCAM Microstructure Classification



Closing

Takeaway

End with the decision, ask, or technical conclusion you want the audience to remember.

- **Deliverable:** Name the concrete output.
- **Primary risk:** Name the hardest unresolved issue.
- **Immediate next step:** Name the action that starts now.

